

WASP-11b

R-1290

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_200102 · created 2026-07-31T16:10:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458850.6947 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.029
Transit depth (Rp/Rs)^2	3.12 +/- 1.04 [%]
Orbital Inclination (inc)	88.42 +/- 1.13
Airmass coefficient 1 (a1)	0.884 +/- 0.078
Airmass coefficient 2 (a2)	0.1 +/- 0.12
Scatter in the residuals of the lightcurve fit is	8.48 %
Best Comparison Star	#2 - [423, 311]
Optimal Aperture	7.51
Optimal Annulus	9.39
Transit Duration (day)	0.1047 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.029 vs 0.131 (deviation 1.22σ)
Duration fit vs published	0.1047 d vs 0.1075 d (deviation 0.26σ)
Mid-time O–C	3.9 min
Depth SNR	4.7
Residual scatter	8.48 %

Light curve

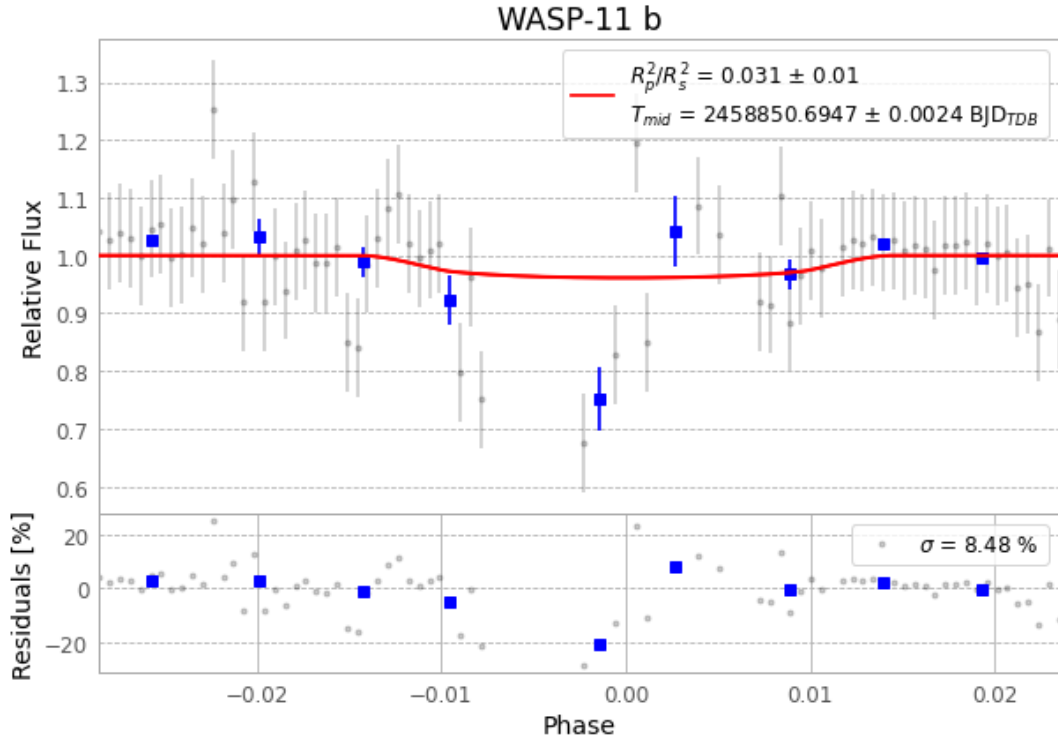


Figure 1 — FinalLightCurve_WASP-11 b_2020-01-01.png (SHA-256 0a6c5fd13f28cf66...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a4bc9bb74704824d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), HATP-10200102020012.FITS → HATP-10200102063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-200102.log (481e813c9c49...), FinalLightCurve_WASP-11 b_2020-01-01.png (0a6c5fd13f28...), FinalLightCurve_WASP-11 b_2020-01-01.csv (5d47d8b09e0d...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:10Z.